# The Architecture of Errors: Logarithmic Mode Discovery and Polylogarithmic Intervention Budgets for Long-Context LLM Reliability

**Anonymous Author(s)**
Affiliation
Address
email

## Abstract

Prior work on length-decay benchmarks [Anonymous, 2025] argued that LLM reliability does not decay exponentially with output length because errors concentrate at sparse "key tokens" rather than spreading uniformly. This paper extends that framework along an orthogonal axis: LLM errors are not only sparse, they are *repetitive*. The same handful of failure patterns recur across models, datasets, and domains. Inside the small set of key tokens, a latent fraction $\beta$ produce *hard* failures; those hard failures cluster into a finite catalogue of recurring modes whose count grows slowly with the number of observed failures. Under an empirical postulate of logarithmic mode discovery—calibrated against endpoint counts from three published taxonomies (ErrorAtlas, MWPES, HumanEval-error-categorisation), with a direct subsample-discovery curve still outstanding—the intervention budget required to bound *per-hard-token* residual error scales polylogarithmically in sequence length. Sequence-level reliability targets are strictly tighter and approach full-catalogue coverage as the number of hard decisions grows—an asymmetry we surface explicitly rather than burying. Twelve intervention categories from prior work, plus 28 capability-elimination citations across six axes stratified into Patterns A/B/C (Appendix B), are consistent with a small library on the order of tens of interventions covering the head of the failure distribution in the per-hard-token regime. We analyze the polylog conclusion symbolically under several cluster-count laws (logarithmic, Heaps, saturating; Appendix A) and find it robust across the family, though the specific doubly-logarithmic rate is the optimistic special case. Prominent steep-decay counter-evidence [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025], re-audited in Appendix C, decays over variables other than raw token length—compositional graph size, fact count, log-time horizon. This relocates rather than dissolves the practical worry: the framework identifies which interventions help and which do not, but does not claim the regimes where $k_{\text{hard}}$ grows fast are thereby made easy.

## 1 Introduction

The standard worry about long-context autoregressive generation is exponential. If every token has independent error probability $e$, then after $n$ tokens the chance of a fully correct output is $(1-e)^n$, which collapses to zero for any nontrivial $e$ and large enough $n$ [LeCun, 2023, Dziri et al., 2023]. Earlier work in this series [Anonymous, 2025] argued that this worry is misplaced because errors are not distributed uniformly across tokens: only 5–10% of tokens are "key"—genuinely

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dependent on long-range context [Fang et al., 2025]—while the remaining majority become nearly deterministic once enough context accumulates. Replacing $(1-e)^n$ with the two-rate model $P(\text{correct}) = (1-e_{\text{key}})^k(1-e_{\text{non}})^{n-k}$ with $k \ll n$ scaling sublinearly in $n$ recovers the observed long-context coherence and converts the reliability question from "how does $n$ grow?" to "how does $k$ grow?".

**From sparse tokens to recurring patterns.** This paper takes the next step. Sparsity tells us *where* errors live; the natural follow-up is *what* they are. Recent failure-mode atlases supply a striking empirical answer: errors are not only sparse, they are also *repetitive*. ErrorAtlas [Ashury-Tahan et al., 2026], covering 83 models across 35 datasets, organises observed failures into 17 named categories sorted by prevalence, with a long-tailed distribution heavily concentrated in the head. In code, two error types (`AssertionError` and `NameError`) cover 86.35% of failures on HumanEval across 14 LLMs, replicated across 23 models on HumanEval Pro and MBPP Pro [Wen et al., 2024]. In math, MWPES-300K [Sun et al., 2025] categorises 304,865 errors from 15 LLMs across four math word-problem datasets and reports that dataset characteristics shape error patterns systematically. The same picture appears in multi-hop QA [Zhang et al., 2026], agentic tool use [Cemri et al., 2025], and retrieval-augmented generation [Wood and Forbes, 2024]. These observations suggest a third layer in the reliability architecture, complementing key-token sparsity (Layer 1) and within-key stratified-manifold structure (Layer 2 in Anonymous 2025). Within the small set of key tokens, only a fraction $\beta$ produce *hard* failures, and those hard failures cluster into a finite catalogue of recurring modes whose size grows much more slowly than the number of observed events.

**Contributions.** The central result is the conditional polylog intervention budget of Proposition 1: $m \geq \lceil |C|^{1-\varepsilon/e_{\text{hard}}} \rceil$, with a doubly-logarithmic rate as the optimistic special case under the postulate above. The *sequence*-level analogue is strictly tighter and approaches full-catalogue coverage as $k$ grows—an asymmetry we surface rather than bury. Around this central claim the paper does four supporting things. The three-layer formal framework—sparsity ($\alpha$), stratification ($\beta$), and bounded mode catalogue ($|C|$)—partitions key tokens into easy and hard subsets, then adds the recurring-pattern catalogue (§3). The empirical postulate of logarithmic mode discovery is calibrated against ErrorAtlas, HumanEval, and MWPES, with $\sigma \in [0.87, 1.85]$ across anchors and $\sigma \approx 1.85$ adopted from ErrorAtlas as the conservative target. Section 4 synthesises evidence for clustering, cluster-selective interventions, and sublinear length scaling, drawing on $\sim$60 prior published results including the six-axis capability-elimination harvest of 28 quantitatively-anchored citations stratified into Patterns A/B/C (Appendix B). And the re-audit of the most-cited steep-decay results (Appendix C) shows each decays over a variable other than raw token length.

## 2 Related Work

**Failure-mode taxonomies.** ErrorAtlas [Ashury-Tahan et al., 2026], MWPES-300K [Sun et al., 2025], the HumanEval categorisation of Wen et al. [2024], and the RFMDataset [Guo et al., 2025] together establish that, at any given corpus size, the number of distinct named failure modes is small (typically 8–20) and the cumulative coverage of the top modes is high. Domain-specific taxonomies for multi-hop QA [Zhang et al., 2026], multi-agent systems [Cemri et al., 2025], and tool-augmented agents [Yao et al., 2024] report the same pattern.

**Targeted interventions.** Each named cluster has been the subject of focused intervention research. Python execution closes most of the arithmetic-error cluster [Gao et al., 2023a, Chen et al., 2023, Gou et al., 2024b]; constrained decoding eliminates format violations [Suresh et al., 2025, Wang et al., 2025b, Dong et al., 2025]; execution feedback closes most code-logic errors [Shinn et al., 2023, Huang et al., 2023]; process reward models reduce step-level reasoning errors [Wang et al., 2024b, Lightman et al., 2024]; RAG strongly reduces hallucinations [Wood and Forbes, 2024]; preference optimisation closes over-refusal [Karaman et al., 2024]; structured uncertainty fixes most tool-call failures [Suri et al., 2025]. Two structural observations recur: each intervention is *cluster-selective* (residuals belong to a different named cluster, not to the targeted one), and *additivity is approximate but not perfect* [Patel et al., 2026, Le, 2026].

**Length-decay benchmarks.** A parallel literature measures the *shape* of the reliability decay curve. Mild-decay results—Loong [Wang et al., 2024a], GSM-$\infty$ [Zhou et al., 2025], RULER [Hsieh et al., 2024], anchor-based LLMs [Pang et al., 2024]—report log-linear, sigmoidal, or threshold-like decay inconsistent with smooth $(1-\varepsilon)^n$. Steep-decay results [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026] are sometimes cited as exponential-compounding evidence. The apparent tension dissolves on careful reading: every steep-decay paper decays over a variable distinct from raw token length, which Appendix C documents.

**Gap.** What is missing is a quantitative bridge between the small, repeating taxonomy catalogue and a polylog intervention budget. We provide that bridge in §3.

# 3 Theoretical Framework

**Four levels of $|C|$.** Before any proposition, a definitional setup. LLM error analysis routinely conflates four objects that the rest of this paper needs to keep apart:

- **L1: failure events.** Concrete LLM errors observed in a benchmark—the raw data behind any taxonomy.
- **L2: empirical taxonomy categories.** Researcher-chosen labels grouping L1 events. ErrorAtlas's 17 categories, MWPES's top-4 classes, and HumanEval's `AssertionError`/`NameError` partition all live here. L2 is observable but depends on the taxonomer's resolution choices.
- **L3: latent failure modes.** The unobserved underlying clusters that L2 approximates. L3 is the quantity Postulate 1 is morally about, but it cannot be measured directly; we treat L2 as a noisy proxy for L3.
- **L4: capability axes / interventions.** The engineering unit: a Python interpreter, a constrained decoder, a retrieval-augmented generator. L4 is coarser than L2 in practice—one capability axis typically targets several L2 categories at once (§4, Claim B).

Throughout this section, $|C|$ refers to the **L2** count: that is what published taxonomies report. Postulate 1 is therefore *engineering*-relevant because L4 is coarser than L2 (so a small intervention library can sweep across many categories at once), and *epistemically* conditional because L2 is a noisy proxy for the actual L3 mode catalogue whose faithfulness has not been independently measured. Tagging the formal claims at the right level keeps the framework honest where casual error-talk slips.

## 3.1 $\beta$-stratification of key tokens

The two-rate model of Anonymous [2025] distinguishes $k$ key tokens (error rate $e_{\text{key}}$) from $n-k$ non-key tokens ($e_{\text{non}} \ll e_{\text{key}}$). Empirical atlases suggest that even within the key-token class errors are not uniformly distributed: most key tokens are "decisions" for which the model has stable representations; only a fraction concentrate the actual failures.

**Definition 1** (Hard fraction)**.** Partition the $k$ key tokens of a sequence into easy and hard subsets,

$$k_{\text{hard}} = \beta k, \qquad k_{\text{easy}} = (1-\beta)k, \qquad \beta \in (0,1),$$

where hard key tokens have an elevated error rate $e_{\text{hard}}$ corresponding to manifold-transition decisions in the sense of Anonymous [2025, §3.2], and easy key tokens have $e_{\text{easy}} \approx e_{\text{non}}$.

Under Definition 1, the composed sequence-level reliability becomes

$$P(\text{correct}) \;=\; (1-e_{\text{hard}})^{\beta k}\,(1-e_{\text{easy}})^{(1-\beta)k}\,(1-e_{\text{non}})^{n-k}. \tag{1}$$

We do not assume iid token errors: the three rates summarise *conditional per-decision failure hazards* after conditioning on prior decisions being correct, and grouping by stratum yields the multiplicative survival expression. Because $e_{\text{easy}} \approx e_{\text{non}}$ once context accumulates, the failure rate is dominated by the $\beta k$ hard tokens, and we treat $e_{\text{hard}}$ as the load-bearing quantity. The parameter $\beta$ is *latent*: failure-mode atlases report $\Pr(\text{category} \mid \text{error occurred})$, which does not identify $\Pr(\text{hard} \mid \text{key-token decision})$. We carry $\beta$ symbolically.

## 3.2 Empirical postulate: logarithmic mode discovery

Stratification names a target—the $\beta k$ hard-token decisions—but does not yet tell us how many distinct ways those hard decisions can fail. The empirical answer from Section 1 is that failures repeat: a small number of recurring patterns covers most of the mass. The next question is how the catalogue's size grows with the number of observed failures, because that is the quantity an intervention library has to keep up with.

Two candidates exist in the type–token literature: Heaps' law (power-law, $|C| \approx K \cdot k_{\text{hard}}^b$ with $b \in [0.4, 0.6]$ [Manning et al., 2008]), and logarithmic discovery. **Zipfian rank-frequency does not imply logarithmic growth.** Heaps' law is the correct type–token consequence of Zipf-distributed events, and it is power-law, not logarithmic. We therefore state logarithmic mode discovery as an empirical postulate, defensible by direct measurement of error taxonomies—not as a theorem.

Two sample-size variables matter, and conflating them is the seam that lets reviewers in. Let $h(n) = \beta k(n)$ denote the number of hard-token decisions in a single sequence of length $n$; let $T$ denote the number of observed hard-failure events in a corpus used to discover the domain catalogue. These are not the same object: $h(n)$ controls per-sequence exposure, $T$ controls empirical discovery. We write $C_D$ for the full reachable catalogue inside domain patch $D$, $C_{\text{seen},D}(T)$ for the subset discovered after $T$ sampled failures, and $C_{\text{active},D}(n)$ for the subset a single sequence of length $n$ can activate.

**Postulate 1** (Patch-indexed catalogue discovery)**.** Within a fixed application domain $D$, the number of named recurring failure modes discovered after $T$ sampled hard-failure events is bounded by

$$|C_{\text{seen},D}(T)| \;\le\; \min\bigl(A_D + \sigma_D \ln T,\; |C_D|\bigr), \qquad \sigma_D > 0,\; A_D \ge 0.$$

The cap $|C_D|$ is the domain-imposed ceiling. The postulate concerns catalogue *discovery*, not the number of hard decisions inside a single sequence.

**Assumption 2** (Per-sequence active-mode exposure)**.** For a single sequence of length $n$ in domain $D$,

$$|C_{\text{active},D}(n)| \;\le\; \min\bigl(A'_D + \sigma'_D \ln h(n),\; |C_D|\bigr).$$

Primed constants are distinct from those of Postulate 1: corpus discovery and per-sequence activation need not share the same rate.

The two bounds answer different questions, and the rest of the paper picks the right one at each point. Proposition 1's sequence-length scaling rate is governed by $C_{\text{active},D}(n)$ via Assumption 2: how much of the catalogue can a single sequence touch? Full domain-library budgeting—the practical engineering target for a deployed system—uses $C_D$: how large does the library need to be to cover the domain's reachable failure modes? The cap $|C_D|$ enters both: once enough has been sampled (or activated) to saturate the domain ceiling, further growth stops and the budget becomes a domain-constant function of $|C_D|$ alone.

**Empirical defense.** Three published taxonomies anchor the mode-rate parameter $\sigma$ at endpoint corpus scales. ErrorAtlas at $k_{\text{hard}} \gtrsim 10^4$ failures has $|C| = 17$, giving $\sigma \approx 17/\ln(10^4) \approx 1.85$ under $A = 0$. HumanEval [Wen et al., 2024], on a comparable sample size, has $|C| \in [8, 12]$ and accordingly $\sigma \in [0.87, 1.30]$. MWPES, an order of magnitude larger ($k_{\text{hard}} \approx 3 \times 10^5$), still has $|C| \in [15, 20]$ and yields $\sigma \in [1.2, 1.6]$. Across the three anchors, $\sigma$ falls in the range $[0.87, 1.85]$; we carry $\sigma \approx 1.85$—the ErrorAtlas value, the largest of the three—as a conservative downstream target. Endpoint counts at a single corpus scale are not discovery curves: a subsample-vs-discovered-modes measurement has not been published for any LLM error taxonomy at this writing, and remains the explicit empirical test we flag (§6, Appendix A).

**Capabilities are coarser than error classes.** Postulate 1's $\sigma \approx 1.85$ is calibrated against *named error categories*, but a single capability axis often targets several L2 categories at once. A Python interpreter removes the execution-error component of arithmetic, unit conversion, simple counting, list manipulation, and date arithmetic; a constrained decoder eliminates by construction the format-violation cluster and the structural component of "missing required element." Postulate 1 is therefore a conservative input for the engineering question; the formal coverage problem is weighted set cover, with the ranked-category form of Proposition 1 as a tractable proxy. An alternative power-law (Heaps) formulation and symbolic-form sensitivity analysis across candidate cluster-count laws are given in Appendix A.

**Domain patches cap the catalogue.** A model deployed in a fixed application domain occupies a bounded sub-region of the stratified manifold—a *domain patch* $P_D$ [Anonymous, 2025, Li and Sarwate, 2025]. Empirical support: embedding-space intrinsic dimensions of $\sim$ 5–15 across domains, well below ambient $\sim 10^3$ [Park et al., 2024, Li and Sarwate, 2025]; cross-domain accuracy spreads of 20–70% on a single base model [Wang et al., 2024c]; log-popularity thresholds below which scaling fails [Mallen et al., 2023, Kandpal et al., 2023]. None of these *measure* $|C_D|$ directly; together they motivate the patch-indexed hypothesis. The patch makes logarithmic or saturating discovery dynamics plausible inside fixed domains; $\sigma_D$, $A_D$, $\beta_D$, $|C_D|$ are all domain-indexed.

### 3.3 Coverage by a targeted intervention library

With the catalogue defined, the next question is how much residual error a library of $m$ targeted interventions actually closes off. Given a catalogue of $|C|$ recurring modes, a library of size $m$ covers the $m$ most prevalent modes and leaves the remaining $|C| - m$ uncovered. We adopt the log-coverage form

$$F(m; |C|) \;=\; \min\!\left(1,\; \frac{\ln m}{\ln |C|}\right) \tag{2}$$

as a *postulated* closed expression respecting the $\min(1, \cdot)$ cap. Reading the formula: the first few interventions cover disproportionate mass (the head of the distribution is dense), then incremental coverage falls off as the library saturates the catalogue. For our ErrorAtlas anchor ($|C| = 17, m = 5$), $F(5; 17) \approx 56.8\%$; doubling the library to $m = 10$ pushes it to $F(10; 17) \approx 81.3\%$.

This is an empirical best-fit choice, not derived from Zipf. ErrorAtlas sorts its 17 categories by prevalence but does not publish a single cross-domain cumulative top-$m$ curve at this writing; alternative cumulatives (Zipf–Mandelbrot, truncated power law, saturating exponential) match the same qualitative shape with comparable freedom. Appendix A shows the polylog conclusion of §3.4 survives across the family.

Two caveats matter for what comes next. (a) $F$ counts named L2 categories covered, not L4 capability axes deployed; in real deployments, where one capability often sweeps several L2 categories, residual error at a given $m$ is generally lower than (2) predicts. (b) The form is intended for $m \geq 2$, $|C| \geq 2$; we adopt $F(0; |C|) := 0$ and $F(1; |C|) := p_1$ (the mass of the single most prevalent mode) since the log expression is not anchored at $m = 1$. After deploying the top-$m$ library, the residual per-hard-token error rate is

$$e_{\text{res}}(m) \;=\; \big(1 - F(m; |C|)\big)\, e_{\text{hard}}. \tag{3}$$

### 3.4 Main proposition: conditional polylogarithmic intervention budget

We now compose the three ingredients—$\beta$-stratification (Definition 1), the active-mode bound (Assumption 2), and the log-coverage form (Eq. 2)—into a single statement about how many interventions a deployed system actually needs. For *sequence-length scaling* we use Assumption 2, not Postulate 1: the empirical sparsity $\alpha = k/n \in [0.05, 0.10]$ is a *finite-length* statement, and under Anonymous [2025, §3.1]'s $k \sim \log n$ regime, $\alpha$ decays as $O(\log n/n)$.

**Conditionality.** The bound below is engineering math, not a theorem about LLMs. It holds conditional on three things: (i) the log-coverage form of §3.3 applied on its declared domain $m \geq 2$, $|C_{\text{eff}}| \geq 2$; (ii) a residual target $\varepsilon \in (0, e_{\text{hard}})$ that is non-trivial but attainable; (iii) Assumption 2 for any sequence-length rate claim. A different cumulative fitted to the same ErrorAtlas/HumanEval/MWPES anchors would preserve the qualitative polylog conclusion (Appendix A) while changing the exponent.

**Proposition 1** (Per-Hard-Token Intervention Budget)**.** *Let $\varepsilon \in (0, e_{hard})$ be a target per-hard-token residual error rate for sequences in domain D, with $|C_{eff}| \geq 2$, where*

$$C_{eff} \;=\; \begin{cases} C_{active,D}(n) & \textit{(per-sequence scaling),} \\ C_D & \textit{(full domain-library deployment).} \end{cases}$$

*Under the coverage form of §3.3, the smallest intervention library satisfying $e_{res}(m) \leq \varepsilon$ has size*

$$m \;\geq\; \left\lceil |C_{eff}|^{1-\varepsilon/e_{hard}} \right\rceil . \tag{4}$$

*If $C_{eff} = C_{active,D}(n)$ and $k(n) = \Theta(\log n)$, then in the pre-cap regime*

$$m \;=\; O\big((\log\log n)^{1-\varepsilon/e_{hard}}\big),$$

*i.e., the intervention budget grows doubly logarithmically with sequence length. In the cap regime, or for full domain-library deployment,* $m \geq \lceil |C_D|^{1-\varepsilon/e_{hard}} \rceil$ *independent of* $n$.

*Proof.* From §3.3, $e_{\text{res}}(m) = (1 - F(m; |C_{\text{eff}}|))\, e_{\text{hard}}$. Requiring $e_{\text{res}}(m) \leq \varepsilon$ gives $F(m; |C_{\text{eff}}|) \geq 1 - \varepsilon/e_{\text{hard}}$. Since $\varepsilon < e_{\text{hard}}$, the required coverage lies below one and the cap in $F$ is inactive; with $F = \ln m / \ln |C_{\text{eff}}|$, we obtain $m \geq |C_{\text{eff}}|^{1-\varepsilon/e_{\text{hard}}}$, and taking the ceiling gives (4). Assumption 2 gives $|C_{\text{active},D}(n)| = O(\log\log n)$ when $k(n) = \Theta(\log n)$, before the cap. After the cap, $|C_{\text{eff}}| = |C_D|$ and the bound is independent of $n$. □

We label this a *Proposition* rather than a *Theorem* to keep its conditional nature visible. The formal content is engineering math, derived from the empirical postulates of §3.2 and §3.3; it formalises a directional intuition rather than deriving a law of LLM reliability. The doubly-logarithmic rate stated in (4) is the *optimistic special case*—per-sequence, pre-cap, under logarithmic mode discovery. Weaker (but still polylog) rates obtain under Heaps (Appendix A); the cap regime is a domain-constant independent of $n$.

**Sequence-level versus per-hard-token bounds.** Proposition 1 bounds the *per-hard-token* residual error. Sequence-level failure probability is strictly stronger. Define the non-hard-token survival factor

$$S_{\text{base}} \;=\; (1 - e_{\text{easy}})^{(1-\beta)k}\,(1 - e_{\text{non}})^{n-k}, \tag{5}$$

so that the composed survival from (1) can be written $P(\text{correct}) = (1 - e_{\text{res}})^{\beta k}\, S_{\text{base}}$. The sequence-level target $P(\text{correct}) \geq 1 - \varepsilon_{\text{seq}}$ then yields three regimes:

**Regime (i):** $S_{\textbf{base}} < 1 - \varepsilon_{\textbf{seq}}$**.** Even at $e_{\text{res}} = 0$ the surviving non-hard-token failure mass exceeds the sequence-level budget. Hard-token interventions alone cannot meet the target; this is the back-door route by which the exponential-in-$n$ concern re-enters, and it should be diagnosed before any catalogue-budgeting exercise.

**Regime (ii):** $S_{\textbf{base}} \geq 1 - \varepsilon_{\textbf{seq}}$**.** Define

$$\tau_{\text{seq}} \;=\; 1 - \left(\frac{1-\varepsilon_{\text{seq}}}{S_{\text{base}}}\right)^{1/(\beta k)}. \tag{6}$$

For $0 < \tau_{\text{seq}} < e_{\text{hard}}$, the sequence target is met whenever $e_{\text{res}}(m) \leq \tau_{\text{seq}}$, so the required library size is

$$m \;\geq\; \left\lceil |C_{\text{eff}}|^{1-\tau_{\text{seq}}/e_{\text{hard}}} \right\rceil.$$

As the target tightens toward the feasibility boundary, $\tau_{\text{seq}} \to 0$ and $m \to |C_{\text{eff}}|$—essentially every named mode must be covered.

**Regime (iii):** $\tau_{\textbf{seq}} \geq e_{\textbf{hard}}$**.** The sequence target is already met at the baseline hard-token rate; no intervention is required.

A common engineering rule of thumb—"fix $\approx 50$ patterns per domain $\to\approx 90\%$ error reduction"—is best understood as a per-hard-token statement; the analogous sequence-level claim requires correspondingly more interventions and tighter coverage of the tail. Production systems with one-shot reliability targets should plan for the stricter regime—and check whether they are in Regime (i), where catalogue-budgeting cannot help at all.

## 4 Empirical Evidence

We collect evidence for three load-bearing claims: errors cluster into a small recurring set (A); each cluster is addressable by one targeted intervention (B); reliability decays sublinearly with output length (C).

**Claim A: Failure-mode clustering.** The strongest single result is ErrorAtlas [Ashury-Tahan et al., 2026]: 83 models × 35 datasets, ≈ 7,000 failures, 17 named categories with a long-tailed, head-concentrated prevalence ordering. Domain-specific Paretos reproduce the shape inside each domain—math (MWPES top-4 categories dominate per model with strong cross-model overlap [Sun et al., 2025]), code (`AssertionError+NameError` = 86.35% on HumanEval, replicated across 23 models [Wen et al., 2024]), multi-hop QA (a single dominant "missing-evidence" mode [Zhang et al., 2026]), agentic tool use (under 20 recurring modes across two studies [Cemri et al., 2025, Yao et al., 2024]), and RAG (a handful of distinguishable hallucination types [Wood and Forbes, 2024]). The catalogue is also *stable* across models: ErrorAtlas reports a fixed category hierarchy across 83 models; RFMDataset [Guo et al., 2025] finds "strikingly similar failure mode distributions" across ten advanced reasoning models; EDIT [Dai et al., 2025] measures a $\approx 4.7\%$ key-step fraction stable across models. Schaeffer et al. [2025] provide the closest theoretical anchor, proving that the observed aggregate power-law scaling of LLM evaluations requires the per-problem success-rate distribution to be heavy-tailed near $p = 0$—structurally related to Postulate 1.

**Claim B: Cluster-selective capability interventions.** A dedicated harvest yields 28 capability-elimination citations across six independent axes (arithmetic, code execution, format/structure, perception/grounding, knowledge/RAG, verification), each axis independently confirmed by between three and nine citations. The full table and the three structural patterns—Pattern A by-construction, B strong-empirical-with-class-shift, C moderate-with-shift—appear in Appendix B.

Arithmetic is the cleanest case: PAL [Gao et al., 2023a] lifts GSM-Hard from 20.1% to 61.5%, and the residuals are problem-comprehension errors rather than arithmetic ones. Format violations admit something stronger still—constrained decoding zeroes invalid-token probability by construction [Suresh et al., 2025, Wang et al., 2025b, Dong et al., 2025]. Code-logic errors yield to execution feedback (Reflexion + AgentCoder push HumanEval pass@1 from 80% to 96.3%, with residuals in spec-misinterpretation [Shinn et al., 2023, Huang et al., 2023]); reasoning steps to process supervision (Math-Shepherd: Mistral-7B MATH 28.6% $\rightarrow$ 43.5% [Wang et al., 2024b]); hallucinations to RAG, where Acurai [Wood and Forbes, 2024] reports 100% elimination on RAGTruth—a benchmark-conditional empirical result, not a general by-construction guarantee. Two more clusters round out the picture: POROver lifts the not-overrefused rate from 57.6% to 82.1% [Karaman et al., 2024], SAGE-Agent raises When2Call accuracy from 36.5% to 65.2% [Suri et al., 2025].

The class-shift signature—post-intervention residuals belong to structurally different classes—is the empirical content of the cluster-orthogonality claim that underwrites Proposition 1's composition step. Capabilities are coarser than error classes: a single Python interpreter removes execution-error components from five named clusters; constrained decoding eliminates format and the structural component of "missing required element" jointly. A negative control sharpens the selectivity claim. DebugBench [Tian et al., 2024] reports that execution feedback works for syntax and reference errors and is explicitly "unhelpful for logic errors"—exactly where the framework predicts capability provisioning fails.

**Claim C: Sublinear length scaling.** Direct evidence for sublinear—rather than exponential—decay shows up across very different measurement designs. Loong [Wang et al., 2024a] reports GPT-4o on Chain-of-Reasoning declining $81.6\% \rightarrow 32.9\%$ across the $10\text{K} \rightarrow >200\text{K}$ context bins; log-linear over $\log L$, and decisively non-exponential per-token. GSM-$\infty$ [Zhou et al., 2025] attacks the independent-error model from a different angle: exponentially more inference compute yields only *linear* AUC gains, and DeepSeek-R1 maintains $10\%+$ accuracy at 130 reasoning operations where $(1-\varepsilon)^{130}$ for any reasonable $\varepsilon$ predicts near-zero. Press et al. [2023] find a $\approx 40\%$ compositionality gap that is approximately *constant* across the GPT-3 family—direct evidence against per-step independent compounding, which would predict the gap to grow with chain length.

Structural correlates of the same phenomenon appear elsewhere. Pang et al. [2024]'s Anchor LLMs achieve $\approx 99\%$ K/V reduction with only minor accuracy compromise: the effective number of distinct attention-key vectors a model needs is far smaller than $n$. Think-Prune-Train [Costello et al., 2025] raises Pass@1 while Pass@20 plateaus—the manifold-transition signature. RULER [Hsieh et al., 2024] finds threshold behaviour inconsistent with smooth $(1-\varepsilon)^n$. METR [Kwa et al., 2025] fits logistic-in-log-length, which is mathematically sublinear in length itself.

Self-consistency [Wang et al., 2023] is consistent with but does not *prove* clustered structure: Condorcet aggregation of iid Bernoulli chains can yield similar gain magnitudes, so we treat it as suggestive rather than diagnostic. Prominent steep-decay counter-evidence [Dziri et al., 2023, Kuratov et al., 2024, Wan et al., 2026, Karpinska et al., 2024] decays over variables distinct from raw token length; per-paper re-audits showing the relocation pattern are in Appendix C.

## 5 Practical Implications

**Reliability engineering is local patch coverage.** Within a fixed domain patch, Proposition 1 makes reliability a small-catalogue engineering problem rather than an asymptotic scaling problem. A team should initially budget for a library on the order of tens of interventions, then refine from the local mode-discovery curve $C_{\text{seen},D}(T)$ and the empirical rank-frequency distribution. As a rule of thumb consistent with the head-mass figures in ErrorAtlas, HumanEval, and MWPES, $\approx 50$ named interventions cover the bulk of the per-hard-token failure distribution in many measured domains. This is a planning prior calibrated to current taxonomies, not a universal constant; the local mode-discovery curve sets the actual library size for any given deployment—the same base model in cardiology RAG, legal contract drafting, and code review yields three different libraries because $C_D$, $A_D$, $\sigma_D$ all change with the deployment domain.

**Per-hard-token vs. sequence-level targets matter for SLA design.** Per-hard-token residual error—the right metric for continuous-correction systems—has a polylog budget. Sequence-level failure probability—the right metric for one-shot systems—is strictly stricter and approaches full-catalogue coverage as $k$ grows. Production deployments should design SLAs to match the actual cost structure, not read the per-token result as the production SLA.

**Library design as engineering, not asymptotic theory.** An intervention library can be assembled module-by-module, in any order, as long as modules target distinct failure classes. Layer-separated interventions (constrained decoding + retrieval + process supervision + tool call) compose approximately additively in our data; the Le [2026] caveat applies only to interventions sharing a prompt channel. Library construction is therefore highly parallelisable.

**Capability libraries are coarser than error-class libraries.** Five named math classes (arithmetic, units, counting, list manipulation, date arithmetic) collapse under one Python interpreter; three code classes (SyntaxError, NameError, most TypeError) collapse under one execution-feedback loop; format violations and the structural component of "missing required element"—together more than $20\%$ of ErrorAtlas—collapse under one constrained-decoder deployment. A team budgeting for $\approx 50$ interventions per domain can expect to need fewer than 50 capability-axis interventions while covering $\approx 50$ named clusters. The six axes of Appendix B are closer to the right unit of engineering accounting than the twelve clusters.

## 6 Discussion

**By-construction elimination as a boundary case.** A subset of the capability-elimination harvest deserves separate accounting. Seven of the 28 citations (Appendix B; constrained decoders [Suresh et al., 2025, Dong et al., 2025, Li et al., 2026, Zhang et al., 2023, OpenAI, 2024], proof kernels [Ren et al., 2025], static syntax checks) achieve residual error rate equal to zero *by mathematical construction*, not statistically. Constrained decoders set $P(\text{invalid token}) = 0$ at every generation step; the class of grammar-violating outputs is mathematically empty. In this regime the polylog bound of Proposition 1 is loose—covered clusters contribute exactly zero to residual error, not a small positive quantity. The result strengthens, rather than refutes, the framework. And Pattern A applies precisely to structural/verifiable classes (format, syntax, schema, invalid proofs), which sit at the head of the heavy-tailed cluster frequency distribution: the strongest mechanism lands exactly where the catalogue is densest.

**Counter-evidence relocates, not dissolves.** Five prominent steep-decay papers [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026, Karpinska et al., 2024] are routinely cited as exponential-compounding evidence. Every one decays over a variable distinct from raw token length: compositional graph size [Dziri et al., 2023], number of supporting facts [Kuratov et al., 2024], log-time horizon [Kwa et al., 2025], capacity threshold [Wan et al., 2026], evidence scope [Karpinska et al., 2024]. The apparent rapid decay is, in each case, in a quantity our framework already concentrates the action in ($k_{\text{hard}}$ and $|C|$), not in raw sequence length—a relocation, not a dissolution. These regimes remain hard; the framework's value is directing intervention toward capability provisioning along the actual decay axis rather than toward context-window expansion that does not help. Appendix C walks through each paper.

**Limitations.** Postulate 1 is empirical, not derived; a domain with genuinely Heaps-power-law mode discovery would invalidate the doubly-logarithmic special case while preserving the qualitative polylog conclusion. The coverage form of §3.3 is an empirical best-fit; readers should not interpret the numerical match against anchors as theoretical confirmation. Cluster-orthogonality is approximate; Le [2026]'s caveat on prompt-channel interference tightens the bound in shared-channel settings. Domain narrowness: Postulate 1's empirical anchor rests on three taxonomies (ErrorAtlas, HumanEval, MWPES), all published 2025–2026 and covering general/code/math; the framework is untested on agentic workflows, long-running scientific reasoning, and multi-turn tool use over millions of tokens—precisely the regimes where $|C|$ may grow faster than logarithmically. The $\sigma \approx 1.85$ estimate is a single-point calibration, not a discovery-curve fit; a subsample-vs-distinct-modes plot has not been published for any LLM error taxonomy at this writing. $\beta$ is latent and we report no empirical range. Taxonomy granularity (the L2–L3 gap) is unmeasured. Patch-shift between deployments changes $A_D, \sigma_D, \beta_D, |C_D|$ simultaneously; libraries calibrated against one patch under-cover the next. Finally, the framework *relocates* the difficulty of long-context reliability rather than resolving it: where $k_{\text{hard}}$ grows with task length (adversarial compositional structure, multi-hop chains, long agent horizons), reliability remains hard. The contribution is to name the on-axis intervention, not to make those regimes easy. A falsifiability test the framework passes: DebugBench [Tian et al., 2024] explicitly catalogues which classes execution-feedback can and cannot repair, and the empirical split (works for syntax/reference; "unhelpful for logic errors") is exactly what the framework's selectivity claim predicts.

# 7 Conclusion

LLM reliability is often framed as an asymptotic scaling problem. Prior work in this line [Anonymous, 2025] argued that the framing rests on a false uniformity assumption—errors concentrate at $\approx$ 5–10% of tokens. This paper takes the next step: within that sparse set, errors are not only concentrated but also *repetitive*, clustering into a finite catalogue whose size, under the empirical postulate of §3.2, grows logarithmically—and under the conservative Heaps alternative (Appendix A), as a small power—of observed failures. The conditional consequence (Proposition 1) is that the intervention budget required to bound per-hard-token residual error scales polylogarithmically in sequence length within a fixed domain patch, and becomes a domain-constant once the patch ceiling $|C_D|$ is reached. Sequence-level reliability targets are strictly tighter. Available evidence is consistent with libraries on the order of tens of interventions covering the head of the per-hard-token failure distribution in many fixed domains. The exact number is patch-indexed and should be estimated from local discovery and rank-coverage curves, not assumed from cross-domain priors. This reframes the question from "can we bound the growth of $n$-token error?" to "have we catalogued enough failure modes?" The latter is finite and addressable. Direct empirical measurement of the mode-rate constant $\sigma$ on new domains—agentic, scientific, code-in-production—is the open empirical question this paper invites; future work flagged by Anonymous [2025, §6] on attention-mechanism bounds and stratified-manifold geometry would replace the empirical postulate with a derived result.

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# A Heaps Power-Law Variant and Cluster-Count Sensitivity

A reader who prefers to derive cluster-count growth from standard Heaps' law rather than from Postulate 1 obtains a qualitatively similar result. Let $|C|(k_{\text{hard}}) \approx K \cdot k_{\text{hard}}^b$ with $b \in (0, 1)$. Canonical fits give $b \approx 0.5$ for natural-language vocabularies; failure-mode taxonomies plateau much more sharply (ErrorAtlas stabilises at $|C| = 17$ across $10^4+$ failures), implying a smaller effective $b \approx 0.2$ in our setting. Composing with $k = \Theta(\log n)$, we get $|C| = O((\log n)^b)$, and Proposition 1 becomes

$$m = O\Big((\log n)^{b\cdot(1-\varepsilon/e_{\text{hard}})}\Big).$$

This is still polylogarithmic in $n$ for any $b \in (0, 1)$ and any $\varepsilon < e_{\text{hard}}$. The paper's qualitative claim survives either choice of cluster-count law.

**Symbolic-form sensitivity across candidate laws.** The polylog conclusion depends on which cluster-count law one accepts; available evidence is consistent with multiple candidates because no subsample-discovery curve has been published for any LLM failure-mode taxonomy at this writing. We therefore report symbolic rates rather than fitted constants. With $h(n) = \beta k(n)$:

- **Logarithmic:** $|C_{\text{active},D}(n)| = O(\log h(n))$. If $k(n) = \Theta(\log n)$, then $m = O\big((\log\log n)^{1-\varepsilon/e_{\text{hard}}}\big)$.
- **Heaps:** $|C_{\text{active},D}(n)| = O\big(h(n)^b\big)$ with $b \in (0, 1)$. If $k(n) = \Theta(\log n)$, then $m = O\big((\log n)^{b\,(1-\varepsilon/e_{\text{hard}})}\big)$.

- **Saturating:** $|C_{\text{active},D}(n)| \leq |C_D|$. Then $m = O\big(|C_D|^{1-\varepsilon/e_{\text{hard}}}\big)$, constant in $n$ once the patch ceiling is reached.

The qualitative conclusion is robust **under the** $k(n) = \Theta(\log n)$ **regime**: $m$ grows more slowly than any positive power of $n$ under every candidate, and the directional claim ("a small library covers the head of the failure distribution in the per-hard-token regime") survives. Only the exponent shifts: doubly-logarithmic under logarithmic discovery, $(\log n)^b$ with small $b$ under Heaps, constant in the cap regime. The doubly-logarithmic rate is the optimistic special case. If $k(n)$ grows as a positive power of $n$, the Heaps variant inherits that power and the polylog-in-$n$ language fails along that axis—the framework's intervention prescription still applies, but its asymptotic-rate framing does not.

Numerical constants require a measured discovery curve $C_{\text{seen},D}(T)$ or $C_{\text{active},D}(n)$. Existing taxonomies provide endpoint category counts at a single corpus scale (ErrorAtlas at $|C| = 17$ for $\approx 10^4$ failures), not discovery curves. The subsample-discovery measurement remains the explicit empirical test that would either tighten the postulate or fall back to the Heaps variant. Until that measurement exists, the framework's headline rate should be read as *polylogarithmic in the pre-cap regime, domain-constant in the cap regime*—with the specific exponent flagged as a falsifiability test rather than a fitted prediction.

# B Full Failure-Mode Taxonomy and the Capability-Elimination Harvest

The intervention literature provides at least one targeted countermeasure for each named cluster of §4. Two organising granularities exist: at the capability level (six axes) the cluster-orthogonality property is most clearly visible; at the error-class level (twelve named clusters) the evidence aligns with the taxonomies of §4 but is finer-grained than the underlying capability mechanisms.

**Six capability axes and three structural patterns.** A dedicated harvest yields **28 quantitatively-anchored citations** across **six independent capability axes**: Arithmetic (Python/symbolic execution), Code Execution (REPL/sandbox feedback), Format/Structure (constrained decoding, FSMs, grammar engines), Perception/Grounding (visual grounding for GUI, charts, tables), Knowledge/RAG (dense retrieval and citation grounding), Verification (proof checkers, learned verifiers, classifier rerouting, process supervision). Each axis is independently confirmed by between three and nine citations. We stratify the 28 by kind of evidence into three patterns:

**Pattern A — Hard guarantees (by-construction).** Seven citations achieve residual error rate equal to zero *by construction*, restricted strictly to structural/verifiable classes: constrained decoders set $P(\text{invalid token}) = 0$ at every step [Suresh et al., 2025, Zhang et al., 2023, Dong et al., 2025, Li et al., 2026, OpenAI, 2024]; static syntax checks reject programs with a `SyntaxError` before execution [Wen et al., 2024]; proof kernels reject any output failing type-checking [Ren et al., 2025]. The class of grammar-violating outputs is mathematically empty under these mechanisms.

**Pattern B — Strong empirical reductions with class-shift signature.** Roughly fourteen citations report empirical reductions of 80–100% in a named error class, with the post-intervention failure log dominated by structurally different residual classes. Program-of-Thoughts on GSM8K [Chen et al., 2023]: calculation errors drop from 30% of failures to 0%, residuals are 62% reasoning + 36% misunderstanding. OpenMedCalc [Goodell et al., 2025]: "only interpretation errors were identified." Acurai [Wood and Forbes, 2024]: 100% hallucination elimination on RAGTruth (95% CI 91–100%)—strong empirical, not by-construction.

**Pattern C — Moderate reductions ($60$–$80\%$) with residuals shifting outside the target class.** Legal RAG [Dantart, 2026]: fabricated citations $> 30\% \to < 0.2\%$. GPT-5 SimpleQA with web access [OpenAI, 2025]: $47\% \to 9.6\%$ (inter-condition, not within-condition). CRITIC [Gou et al., 2024a]: toxic generation $-79.2\%$.

**The 28 citations, organised by axis and pattern.**

| Axis | Citation & targeted class | Pre → Post | Pattern |
|---|---|---|---|
| Verification | Ren et al. [2025] — Invalid Lean proof | any → 0% by constr. | A |
| Format | OpenAI [2024] — JSON schema violation | >60% → 0% by constr. | A |
| Format | Zhang et al. [2023] — Tool-call syntax | 21–100% → 0% | A |
| Format | Suresh et al. [2025] — JSON parse failure | 13–82% → 0% | A |
| Format | Dong et al. [2025] — Multi-format errors | 20–38% → 0% | A |
| Format | Li et al. [2026] — Malformed tool calls | 33–78% → 0% | A |
| Arithmetic | Chen et al. [2023] — Calculation on GSM8K | 30% → 0% of failures | B |
| Arithmetic | Goodell et al. [2025] — Clinical arithmetic | "only interpretation errors" | B |
| Knowledge/RAG | Wada et al. [2025] — RAG hallucinations | 8% → 0% ($p = 0.012$) | B |
| Knowledge/RAG | Wood and Forbes [2024] — Context-conflict hallucinations | 100% → 0% on subset | B |
| Knowledge/RAG | Dantart [2026] — Fabricated legal citations | >30% → <0.2% | C |
| Code Exec | Wen et al. [2024] — `SyntaxError` (HumanEval) | 5.76% → 0.01% | A |
| Code Exec | Li et al. [2022] — False-positive submissions | 62% → 4% | B |
| Code Exec | Shi et al. [2024] — 5 code-bug classes | 100% repair (5/6 classes) | B |
| Knowledge/RAG | Gao et al. [2023b] — Citation grounding (ELI5) | ≈50% w/o complete support | C |
| Knowledge/RAG | OpenAI [2025] — Factual errors | 47% → 9.6% w/web | C |
| Knowledge/RAG | Zakka et al. [2024] — Clinical citation errors | 44% → 9% | C |
| Arithmetic | Wang et al. [2025a] — Arithmetic in medical reasoning | 426 → 74 errors | B |
| Code Exec | Wen et al. [2024] — `NameError` repair | 22.7% → 2.3% | C |
| Perception | Gou et al. [2025] — GUI grounding errors | 16.2% → 73.3% acc. | B |
| Perception | Xie et al. [2025] — GUI task failures | 5% → 27% SR (5.4×) | B |
| Perception | Cheng et al. [2024] — Element mislocation | 5.2% → 53.4% (10.3×) | B |
| Perception | Liu et al. [2023] — Chart-reading errors | 38.2% → 67.6% (+29.4 pp) | B |
| Verification | Gou et al. [2024a] — Toxic generation | −79.2% | C |
| Verification | Wang et al. [2024b] — Reasoning step errors | 28.6% → 43.5% w/rerank | B |
| Knowledge/RAG | Asai et al. [2024] — Unsupported facts | 55.5% → 22% error | B |
| Perception | Lu et al. [2024] — Icon/element errors | 70.5% → 93.8% (79% red.) | B |
| Knowledge/RAG | Mallen et al. [2023] — Long-tail entity errors | ≈80% → ≈50% failure | B |

**Capabilities are coarser than error classes.** Several entries in the harvest reveal "two-for-one" reductions where one capability addresses multiple named clusters: a single Python interpreter removes the execution-error component of arithmetic, unit conversion, simple counting, list manipulation, and date arithmetic; constrained decoding eliminates by construction both format violations and the structural component of "missing required element"; code execution feedback strongly reduces `SyntaxError`, `NameError`, and most `TypeError` together; RAG strongly reduces factual hallucinations, fabricated citations, and outdated information jointly. The practical capability library required is therefore smaller than the count of named error categories.

**The twelve named clusters.**

| Cluster | Axis | Intervention | Before → After | Patt. |
|---|---|---|---|---|
| A. Arithmetic | Arithmetic | PAL Python [Gao et al., 2023a] | 20.1% → 61.5% | B |
| B. Unit conversion | Arithmetic | Folded into A (Python) | — | — |
| C. Counting | Arithmetic | Explicit counter (gap) | — | gap |
| D. Format/schema | Format | DINGO/SLOT [Suresh et al., 2025, Wang et al., 2025b] | up to +68 pp; 99.5% acc. | A |
| E. Code logic/type | Code Exec | Reflexion/AgentCoder [Shinn et al., 2023, Huang et al., 2023] | 80% → 96.3% pass@1 | A/B |
| F. Multi-hop drift | Knowledge | Entity-grounded rewriter | ≈5–10 pp | C |
| G. Reasoning step | Verification | Math-Shepherd [Wang et al., 2024b] | 28.6% → 43.5% | B |
| H. Spec misinterpret. | Verification | Clarification [Niwa and Iso, 2024] | ROUGE-L +15 | C |
| I.a Struct. missing | Format | Subsumed by D (constr. decode) | — | A |
| I.b Semantic missing | Verification | Subsumed by G/H | — | C |
| J. Hallucination | Knowledge | RAG [Wood and Forbes, 2024] | non-zero → 0% | B/C |
| K. Refusal | Verification | POROver [Karaman et al., 2024] | 57.6% → 82.1% | B |
| L. Tool/API | Format+Verif. | SAGE-Agent [Suri et al., 2025] | 36.5% → 65.2% | B |

Eight of twelve categories have a strong citation with double-digit percentage-point improvement; three (B, C, I) lack a clean single-cluster ablation; one (F) has consistent medium-strength evidence. Category I (missing required elements) splits mechanistically into I.a (structural, absorbed by D's constrained decoders) and I.b (semantic, absorbed by G/H). Categories B (units) and C (counting) remain technical gaps: B is naturally folded into A; C is the smallest residual gap.

**Additivity and its limits.** Patel et al. [2026] demonstrate that stacking interventions targeting orthogonal failure modes can produce large compound reliability gains: their parallel-consensus framework yields a $14{,}700\times$ improvement over single-pass baseline—evidence for compound gains from decomposition plus consensus aggregation under their specific parallel-voting regime, not direct demonstration that heterogeneous cluster-targeted interventions compose additively without voting. Shang et al. [2024] similarly show supra-additive gains when modules target distinct failure modes. Le [2026] reports that schema-level and prompt-level instructions interact *non-additively* when sharing a prompt channel: interventions on orthogonal processing layers (decoding constraint vs. retrieval vs. training-signal vs. inference-time tool call) compose near-additively, while those sharing a channel may interfere.

**The irreducible-semantic residual.** Of the 17 ErrorAtlas categories, 13 are addressed under Patterns A, B, or C by one of the six capability axes; four are not—residuals of inappropriate refusal beyond preference optimisation, specification misinterpretation not closed by clarification, problem-decomposition reasoning bottlenecks, and a "user wanted something different" semantic remainder. These are classes where the failure is in choosing what to do, not executing it. Proposition 1's prediction of $O(\log)$ rather than $O(0)$ residual reliability reflects exactly this irreducible core. The framework does not claim $100\%$ coverage of all failures; it claims polylog-bounded *capability-eliminable* failures, with the named semantic residual as the floor.

## C Counter-Evidence Re-Audits

Five prominent papers are routinely cited as evidence that LLM reliability decays steeply with length [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026, Karpinska et al., 2024]. A careful re-reading shows that *every one* of these papers decays over a variable distinct from raw token length $n$. Identifying the decay axis is not the same as dissolving the practical concern: where compositional graph size, fact count, or evidence scope grow with problem length, the framework predicts steep failure curves. The contribution is to identify which interventions help (capability provisioning along the actual decay axis) and which do not.

**Dziri et al. [2023], Faith and Fate.** GPT-4 multi-digit multiplication accuracy drops from 59% (3-digit) to 4% (4-digit) zero-shot, with the authors theorising "probability of incorrect predictions converges exponentially to $\approx 1$ for abstract compositional tasks." The decay variable is *compositional graph size* $N$, not raw token length: the $3 \times 3$ graph has on the order of $d^2$ partial products plus carries, and multi-digit multiplication is engineered so that $k_{\text{hard}} \approx N$—every node in the computation graph is a hard decision. For natural-language tasks where $k_{\text{hard}} \ll n$, Dziri's regime is the *boundary case* in which our framework reduces to their result. A clean $(1-\varepsilon)^N$ exponential cannot simultaneously reproduce 59% at 3-digit and 4% at 4-digit for any single per-node $\varepsilon$—the observed drop is locally steeper than per-node iid exponential, consistent with a finite catalogue of failure modes exhausting as $N$ grows. *Where the framework concedes*: in adversarial compositional tasks engineered so $k_{\text{hard}} \approx N$, the framework reduces to Dziri's regime and does not relieve it; the relocation is informational, not magical.

**Kuratov et al. [2024].** The abstract reports "performance declines sharply with increased reasoning complexity" and models "effectively utilise only 10–20% of the context." Read in detail, BABILong varies two axes: context length $n$ (0K to 10M tokens) and number of supporting facts $k$ (QA1 $= 1$ fact to QA3 $= 3$ facts). The sharp decline is in $k$, not $n$. For QA1 (single-fact), most models "perform well up to 4,000 tokens"—a plateau, not exponential decay. The famous RAG-flat-across-length result (60% on single-fact QA *independent of context length*) is the cleanest possible demonstration that when relevant evidence is in window, length does not matter. Recurrent Memory Transformers maintaining performance to 50M tokens further confirms effective $k$ is determined by architecture, not raw $n$. The framework's concession is narrow but real: for multi-hop tasks whose required fact count grows with task complexity, BABILong's sharp $k$-axis decay is exactly what the framework predicts happens, not a counter-example.

**Kwa et al. [2025] (METR).** Per-model success fits a logistic in $\log$(human-task-duration): $S(t) = \sigma(\beta(\log t - \log h))$. Logistic-in-log-length is *mathematically sublinear* in length itself. The 80%-horizon being $4$–$6\times$ shorter than the 50%-horizon is a steep within-model cliff but compatible by construction with the polylog result—a cliff at a specific capacity threshold is a manifold-transition signature. The famous exponential—capability doubling every seven months—is an *inter-model* claim about how the horizon $h$ moves across generations, orthogonal to within-model decay shape. An honest qualifier: the within-model logistic-in-log cliff is steep on a practical scale. Calling it "sublinear in length" is technically correct but engineering-useful only for systems designed to operate well below the 50% horizon.

**Wan et al. [2026], Fano-style upper bound.** This paper theorises super-linear information-demand growth and identifies an "accuracy cliff" at capacity overflow: "when the task's information demand surpasses the model's output capacity, performance does not degrade gracefully but instead collapses sharply." The behavioural prediction is a *threshold*, not smooth $(1-\varepsilon)^n$. A cliff at a specific capacity threshold is exactly the manifold-transition behaviour our framework predicts at the boundary between covered and uncovered modes; Wan et al.'s mechanism (capacity overflow) is one specific cause, complementary to ours. *Where the framework concedes*: Wan documents a mechanism by which $|C|$ effectively exceeds the model's coverage in a single forward pass; Pattern A interventions (constrained decoding, formal verification) are the kind of response the framework expects but does not automatically supply.

**Karpinska et al. [2024].** GPT-4o achieves 55.8% pair accuracy across 1,001 minimally-different true/false claim pairs about long fictional books (mean length 127K tokens). The paper reports performance by *evidence scope*, not by context length: 59.8% on sentence-level retrieval, 47.6% on passage-level, 41.6% on global reasoning. No per-context-length curve is reported. The decay axis is the number of evidence pieces that must be integrated—a $k$-axis quantity, not an $n$-axis one. NoCha is therefore not counter-evidence to a sublinear-in-$n$ claim. Evidence-scope decay is exactly the $k$-axis observation the framework predicts cannot be addressed by scaling raw context length—it requires retrieval or process supervision along the actual decay axis. The relocation is informational, not magical.

**Unifying observation.** Every counter-paper decays over a variable that is not $n$: compositional graph size, fact count, log-time horizon, capacity threshold, or evidence scope. The apparent rapid decay is, in each case, in a quantity our framework already concentrates the action in ($k_{\text{hard}}$ and $|C|$), not in raw sequence length. This is a relocation, not a dissolution. Where $k_{\text{hard}}$ grows with task length—adversarial compositional structure, multi-hop fact chains, long horizons that force more decisions—reliability remains hard; the framework's value is directing intervention toward capability provisioning along the actual decay axis rather than toward context-window or compute-budget expansion that does not help.

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